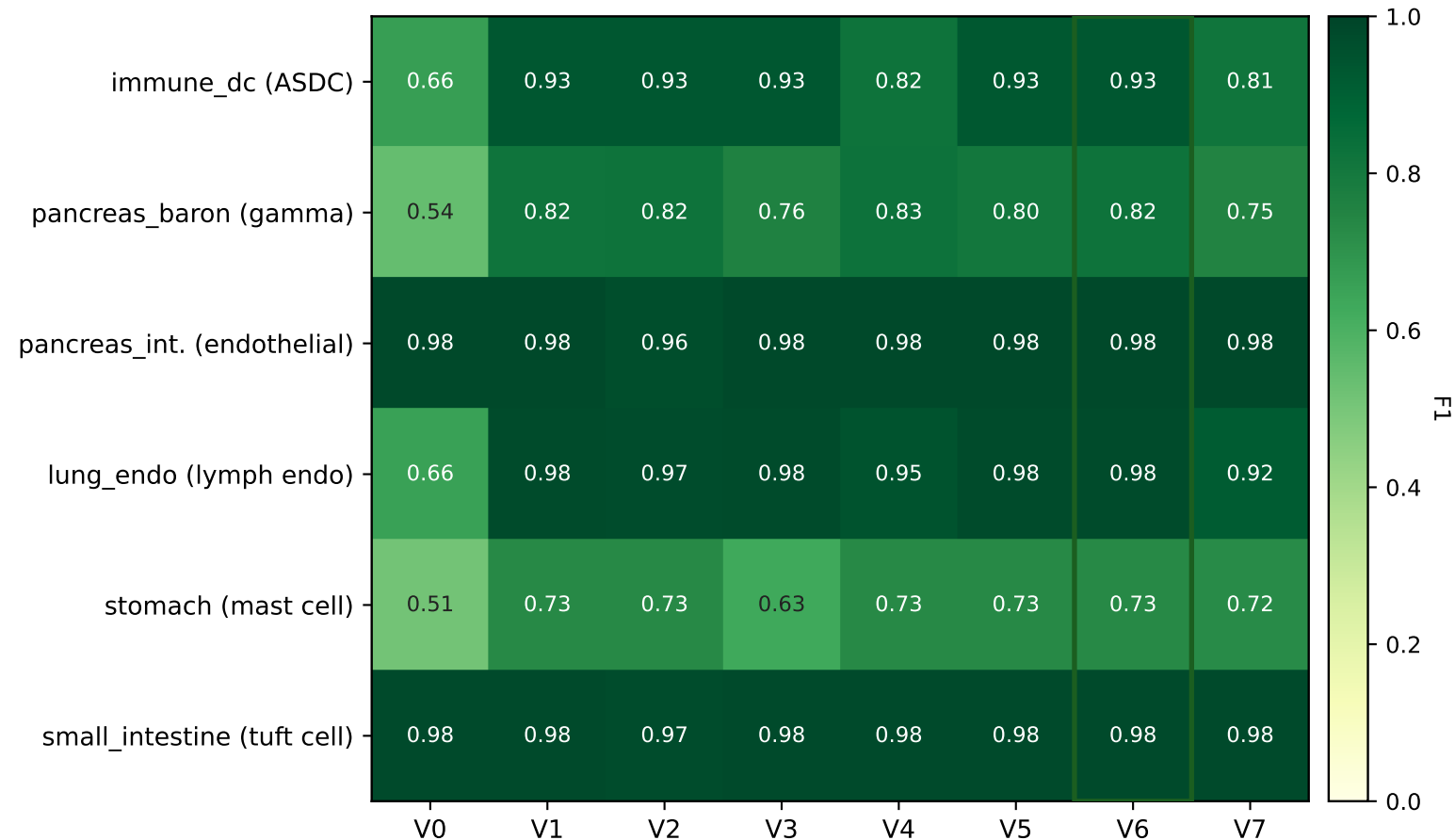
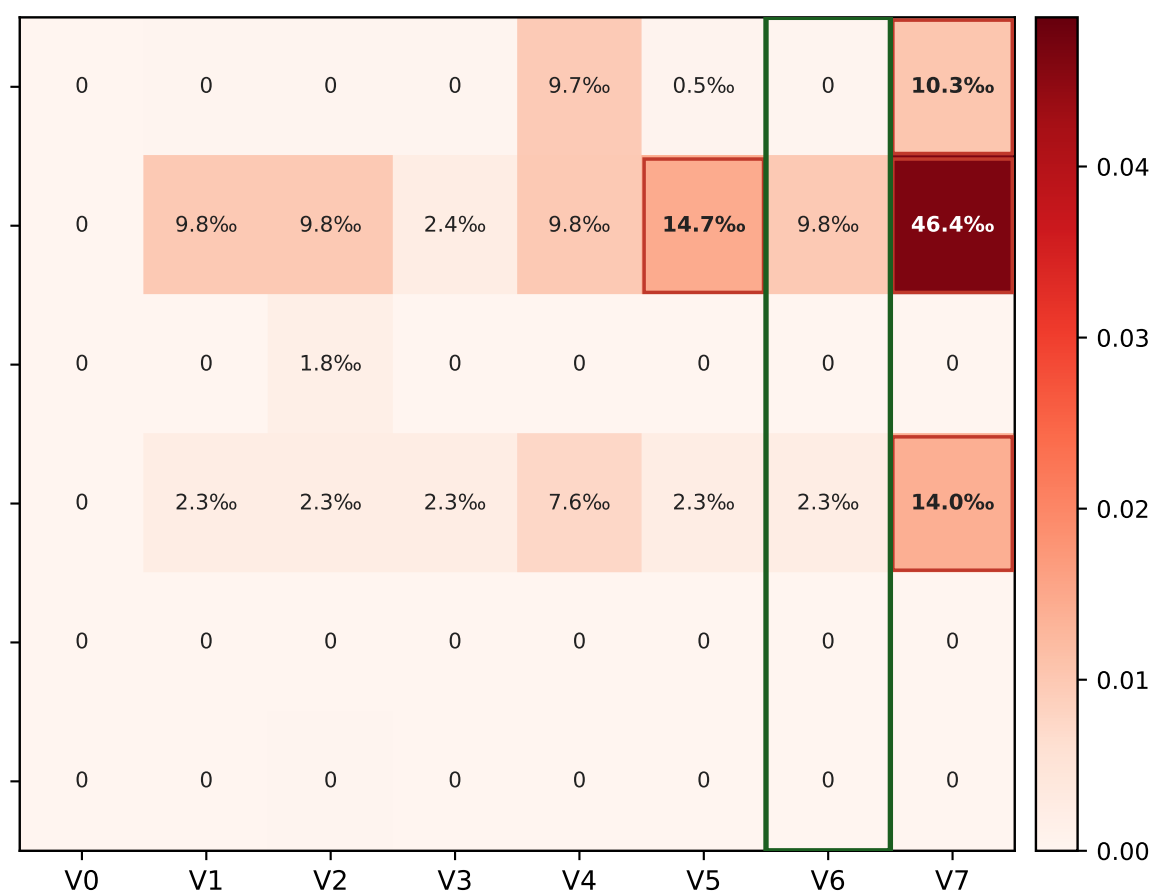


Ablation matrix · 6 datasets × 8 variants · seed=42

rare F1 (mean over 4 rts)



FFR (max over 4 rts) · red dashed = $\alpha=0.01$ contour



V0=baseline scANVI V1=no sep gate (LOW_SEP=0) V2=no necessity guard V3=rank=1 fixed V4=rank=2 fixed V5=no conformal τ V6=full (ours) V7=rank=3 fixed (sensitivity)

Green outline = full method (V6, ours). Red outline = FFR exceeds $\alpha=0.01$.